

Comparative Study of REINFORCE and A2C Variants

1. Introduction

This report focuses on **comparing REINFORCE and three variants of Advantage Actor-Critic (A2C)**:

- **A2C Separate**: distinct actor and critic networks
- **A2C Shared**: shared encoder with two output heads
- **A2C Shared + GAE**: shared network with Generalized Advantage Estimation

The objective is to evaluate how **network architecture** and **advantage estimation techniques** affect learning performance, stability, and convergence speed across environments of increasing complexity:

1. **Classic Control**: Pendulum-v1
2. **Box2D**: LunarLanderContinuous-v2
3. **Atari**: Adventure-v5

2. Methodology

1. Algorithms

We implemented four policy-gradient reinforcement learning algorithms:

1. **REINFORCE (Monte-Carlo Policy Gradient)**: A baseline algorithm that updates the policy using the full-episode return.
2. **A2C (Advantage Actor-Critic) with Separate Networks**: Actor and critic are parameterized by independent networks allowing specialized feature representations for policy and value estimation.
3. **A2C with Shared Network**: Actor and critic share a common encoder followed by separate output heads for policy and value, promoting parameter efficiency and shared feature learning.
4. **A2C with Generalized Advantage Estimation (GAE)**: Same architecture as shared A2C but incorporates $GAE(\lambda)$ to reduce

variance in advantage estimates while maintaining low bias.

2. Network Architecture

- **REINFORCE Policy Network:** Two fully connected layers with 128 units and ReLU activations, Gaussian policy with learnable log-standard deviation.
- **A2C Separate Networks:** Actor and critic each have two fully connected layers (128 units, ReLU), actor outputs mean and log-standard deviation.
- **A2C Shared Network:** Shared 128-unit encoder with ReLU, separate linear heads for actor and critic outputs, actor outputs mean and log-standard deviation.

3. Hyperparameters

Hyper-parameter	Discount factor (γ)	lr	Entropy coefficient	GAE λ	Max episodes	Random seed
value	0.99	3e-4	0.01	0.95	20,000	42

All experiments were run with identical hyperparameters to ensure comparability.

4. Experimental Protocol

1. **Environments:** Selected Pendulum-v1 from Classic Control, LunarLanderContinuous-v2 from Box2D and Adventure-v5 from Atari
2. **Seeding:** Environment and PyTorch seeds set for reproducibility.
3. **Training:** Each algorithm was trained for a fixed number of episodes, logging per-episode rewards.
4. **Evaluation:** Performance was measured using average return, standard deviation and convergence speed.
5. **Visualization:** Reward curves were plotted to compare learning stability and sample efficiency across algorithms.

3. Experimental Setup

- **Software:** Python, Gym, Pandas, Matplotlib, PyTorch.
- **Hardware:** CPU for Classic Control Env and GPU for the others.
- **CSV files:** included episode number, reward, time per episode.

4. Results

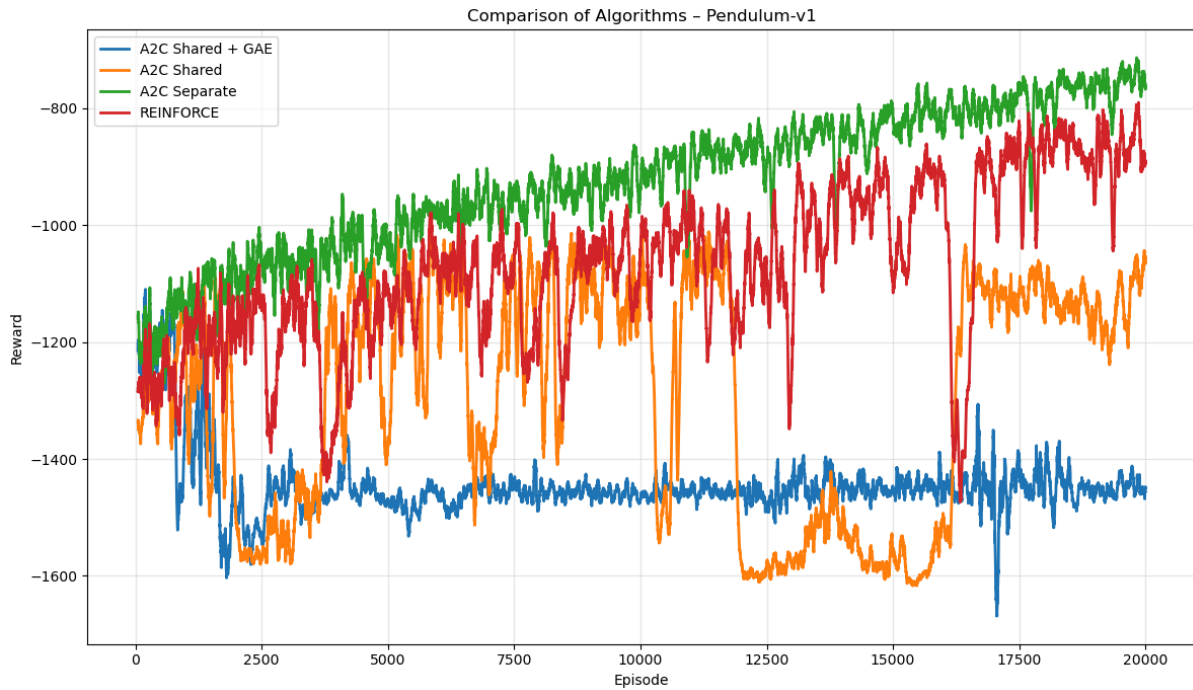
1. Performance Table

Environment	Algorithm	Average Return	Std Return	Convergence Episode
Classic Control	A2C shared + GAE	-1444.48	139.60	50
	A2C shared	-1289.33	247.52	1341
	A2C separate	-926.93	184.06	13039
	Reinforce	-1069.15	219.63	13106
Box2D	A2C separate	75.91	120.42	11146
	Reinforce	99.17	101.14	9226
	A2C shared + GAE	-52.27	89.66	6463
	A2C shared	-8.29	102.84	15221

2. Observations from plots

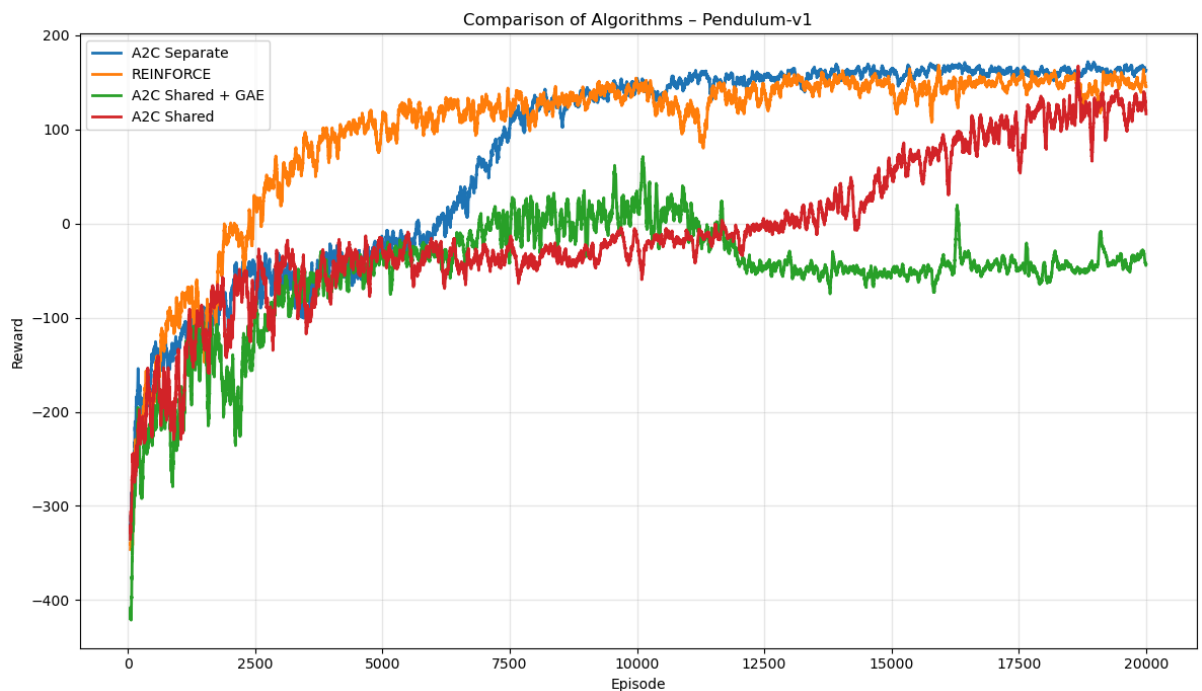
Classic Control Env:

- A2C Shared + GAE converges fastest and maintains stable learning.
- REINFORCE and A2C Separate are slower with higher variance.
- A2C Shared without GAE improves over REINFORCE but is less stable than with GAE.



Box2D Env:

- A2C Shared + GAE converges fastest though average reward is lower due to unstable exploration.
- REINFORCE performs reasonably well achieving high average return but slower convergence than A2C Shared + GAE.
- A2C Shared alone is slower and less stable, showing high variance.



5. Analysis and Discussion

- **Impact of GAE:** significantly reduces variance in A2C, leading to faster and more stable convergence in both environments.
- **Network sharing:** shared encoder reduces parameter count but can increase interference, combining shared network with GAE yields best stability.
- **REINFORCE:** simple to implement but high variance leads to slow convergence and unstable learning, especially on complex environments.
- **Environment complexity:**
 - Pendulum-v1 (simple, continuous, low-dimensional) shows clearer trends.
 - LunarLanderContinuous-v2 is higher-dimensional and more stochastic, algorithms are more unstable and requiring more episodes for convergence.

Key Insights:

1. **Best performing algorithm:** A2C Shared + GAE for both environments in terms of convergence speed and stability.
2. **Trade-offs:** simpler algorithms like REINFORCE are more sensitive to variance and may require significantly more episodes.
3. **Architectural choices:** shared vs separate networks affect convergence patterns, GAE is crucial for stability.

6. Atari Environment (ALE/Adventure-v5)

The Atari environment (Adventure-v5) was included to evaluate algorithm performance on high-dimensional, image-based state spaces where convolutional encoders are required to extract features from raw pixel observations. However, due to hardware constraints and long training times, full experimental results could not be obtained this week.

Methodology Differences

While the underlying algorithms (REINFORCE, A2C separate, A2C shared, A2C + GAE) remained the same, the network architecture and input preprocessing differed from the Classic Control and Box2D environments:

- **Input:** RGB frames (shape $[C,H,W]$), normalized to $[0,1]$.
- **Feature Extraction:** Convolutional Neural Network (CNN) encoder with three convolutional layers followed by fully connected layers to produce compact feature vectors.
- **Policy / Value Heads:**
 - **REINFORCE:** FC layers output action logits.
 - **A2C Separate:** Actor and critic each have their own CNN encoder + head.
 - **A2C Shared:** Single CNN encoder feeding both actor and critic heads.

Experimental Setup Differences

Aspect	Classic/Box2D	Atari
State	Low dimensional vector	High dimensional image (84×84×3)
Network	Fully connected layers	CNN encoder + FC layers
Episodes	20,000	10 (because of the GPU availability)
Training time	4h/14h	~40min on GPU
Device	CPU/GPU	GPU

Due to Kaggle's GPU limits (30 hours/week) and prior usage, further training could not be performed this week.

Results

Training for 10 episodes resulted in **0 reward** for all algorithms, which is expected given the complexity of the environment and the extremely limited number of episodes. This demonstrates that Atari

environments require substantially longer training and more computational resources to observe meaningful learning.

